# Automated File Storage and Retrieval System with Robotic Arm for Cloud-Based Printing

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***Abstract*— This study covers design and development of an initial model of a file storage and compartment in a subscriber-based remote printing system that uses cloud to accept files, temporary stores the print-out in an assigned compartment and releases the corresponding printout upon user arrival. This framework prints the document remotely, potentially minimizing queues to improve the existing Print Agent on Cloud. It consists of a storage system, relay and sensor circuit and a robotic arm controlled by ATMEGA328P-PU microcontroller. The arm, with two linear and two revolute axes, delivers the printout to an assigned compartment for temporary storage until the user arrives for pick-up. The duration of every motor movement is compared where no actual motion is observed to compute for the total system moving efficiency. The initial model successfully receives, prints, stores and releases the file. The time it takes to reserve a compartment is 103 seconds with a moving efficiency of 57.48%. While the total printing time greatly varies based on the number of pages, the time it takes for user pickup after authentication is 183 seconds at an efficiency of 89.07%. Future works may be done to improve robotic arm movement and reduce customer waiting time.**

***Keywords—, Remote Printing, Robotic Arm, Cloud Printing, Print Agent on Cloud***

## I. INTRODUCTION

Printers are very important peripheral devices used to produce hardcopies of file objects. They are found in offices and many households. In this period where electronic copies can be accessed by many devices and file transfer is common through email or other platforms, documents on paper are still in demand for many reasons especially in many students [1]. For personal or academic use, there are still many instances that files are printed in the outside printing facilities. Reasons can be an immediate need for printing or printing a file and in some instances, doing it outside is more practical and convenient than buying a printer. In the case of some megacities like Manila, Philippines, long lines of queues are inevitable. At peak seasons (final weeks or quarter for students), completion of printing service would take so much time. Oftentimes, the huge demand for printing overwhelms the administrators of computer shops and photocopiers. If students need to fall in line and wait for their turn, they may not be able to meet their goal, which is to beat the deadline. There were local attempts to automate the vendo printing process, but are limited to onsite printing where a USB storage device is inserted on a PC and printing service will be activated upon paying the right amount. Some made the printer portable. As an IoT device, the printer can accept data transmission through different network protocols and make print jobs [2]. Also, there are institutions that use centralized wireless printing which adopts Print Agent on Cloud [3]. Any device can send files, the user logs in upon arrival on site before it prints the documents. While these mechanisms offer alternative options, queues are not lessened because it has the same sequence of how the user can print in a computer shop. The only difference is that file sending and payment gets automated.

The demand for document printing increases when the country experience the global COVID-19 Pandemic. This is due to the increased implementation of module-based learning, which provides a way for elementary and high school students to learn without the need for electronic gadgets and internet connection. Modules are printed by the teachers and are distributed to the parents or guardians. Post-pandemic, the module-based learning remains as it offers flexibility and equitable learning.

This study focuses on the design and development of the file storage and robotic arm model as a part of Remote Printing Systems with the aim of providing a framework for contactless service that may eventually speed up the process through automation.

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## II. RELATED LITERATURE

### A. Remote Printing

Because of technology that enables people to do many tasks anytime, anywhere, the level of connectivity affects the changing landscape in many areas. A study by International Data Corporation [4] suggests that 72.3% of the US workforce will be mobile by 2020. Among the main factors causing the increasing mobility include the acceptance of companies of "Bring Your Own Device" (BYOD) programs, mobile devices being more available and cheaper and technologies such as fingerprint scanner, Near-field Communications (NFC) and Augmented Reality (AR). These may contribute to the delivery of files through cables and converted to hardcopies in the right destination. Remote printing contributes to the increasing level of connectivity. In fact, businesses engage remote printing as they move towards the following trends [5].

1. On-premise business apps moving to 3rd party cloud service providers
2. Real-time information exchange between customers and suppliers displacing batch exchange of the same information
3. Increases in the "virtual workforce" at the expense of "on premise" workers

Such enablers benefit the companies adapting remote printing to reduce operational costs and increase profits.

### B. Centralized Printing Schemes

The idea of a centralized printing service in the academe is not new in some Universities of developed countries. It has been conceptualized specially among educational institutions in the United Kingdom. They adopt a specific kind of printing called as the Print Agent on Cloud. It is a popular kind of centralized printing where the vendors provide interface to access the printers that are already pre-registered in the cloud. Those who chose to avail of the service may send files, where it is queued in the cloud. Once the subscriber arrives and authenticates at one of the booths, the file shall be retrieved from the cloud and printed. One disadvantage, however, is that it might also lead to printer congestion if much retrieval happens at the booth at the same time [3].

The University of Sunderland was able implement printing without human assistance through PrintOnLine. It can provide printing service based on topped-up accounts. The printing stations are located inside their libraries and for use of their students only credit [6]. The University of Cambridge, also in United Kingdom, presented a more comprehensive approach in providing printing service to the students using Mobility Print Feature of DS-Print Service. These are installed in the common workstations located across Cambridge campuses and colleges that can be used to print documents. It can accept documents sent through mobile devices and desktops with minimal configuration from smartphones, tablets, laptops and any other compatible device. All these services were accessed through Cambridge Desktop Services account for authentication and common balance [7]. Another school in UK offering similar services is the University of Westminster in London. They also have personalized accounts with balance included in the Identification Cards that enables the Sunderland WebPrint service touched in the university's Kyocera printers [8].

The automated printing operation of the three universities is exclusive to the students, and sometimes, the teacher and staff subscribed in the system. Using the school portal or website, students can upload print jobs from their desktops, laptops and mobile devices to a designated workstation connected to a printer or sometimes already directed to a printing facility, typically a multi-function printer. The University of Westminster, for instance, can store the print job within 24 hours. Subscribers will then have to swipe ID. The printing facility will then check if there is an active print job and if there is enough balance. If these two conditions are satisfied, it will print and charge the subscriber. The university's WebPrint service is also extended to onsite photocopying services. Printing is also available through computers in the University that students can access. Printers[8].

On a bigger perspective, a way of making centralized printing efficient is the use of print management solutions. One of it is the Equitrac Print Management Software. Designed for offices and institutions, the solution is an organized way of auditing print jobs, tracking print patterns and maintains security and responsibility among the users. It claims to have significantly reduced the long-term cost of printing and minimizing wasted prints. When Deer Park Union Free School District, consisting of five schools, in Babylon, New York adopted the solution through multifunction printers, it is able to save $37K on printing costs [8]. Another wireless printing solution is the EFI™ PrintMe® Cloud Service. It claims to provide an easier way of printing and lower IT costs (EFI). Another similar technology is PrinterOn Mobile Printing Solutions. It offers a wide range of cloud printing services from enteprise, public and to private printing [10].

The modern printing system adopted by the educational institutions and the managed services provided by different vendors present an efficient way to centralize printing services across their respective markets. However, there are some limitations in the operation. First, for the schools, their printing facilities will not be able print unless the prepaid card will be swiped. While this may be possible for system with limited subscribers, it would be a problem as the number of subscribers increase. Everyone agrees that printing take time and if the public will just apply what is being done in their universities, it will just be the same as printing inside a computer shop. Customers will just have to wait until all of the preceding print jobs are finished. Furthermore, customers will not determine if their files can be printed smoothly. It will be very disappointing

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to send remote print job miles away only to find out that the computer connected to the printer cannot read the document. Solutions deployed at institutions like universities require pre-validation through card or a password to activate printing.

## III. METHODOLOGY

This section covers the methods of design and system evaluation of the file storage system and robotic arm that can be applied in different types of remote printing systems. For the purpose of providing an initial demonstration of the model, the storage bin will only have five compartments, meaning five print-outs will only be stored at the same time.

### *A. OPERATION*

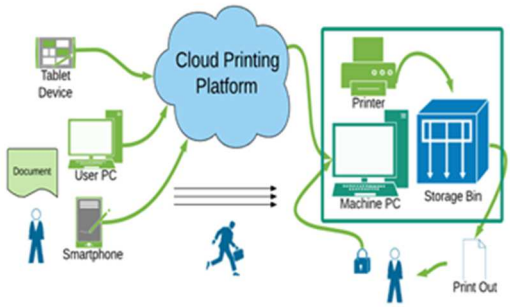



Fig. 1. System Operation

Fig. 1 shows the system operation. It includes the following steps:

1. PRINT. The user, through any internet-connected device, access the system-designated website, logs in and uploads the file for printing. In between this process, the system temporarily assigns a dedicated compartment for the user. A robotic arm will deliver the print-out to the compartment. The system records the number of print job pages. If all the slots are taken by the other users, this step will not be available anymore unless a print-out is retrieved thus, making a compartment available again.

2. GO. The user may arrive on the site while the print-out is temporarily stored in the assigned compartment. Storage time is recorded. The system may also accommodate remote printing and site retrieval (no. 3) for other users.

3. GET. The user arrives on the site, logs in and robotic arm will deliver the print-out for retrieval.

### *B. SYSTEM DESIGN*

Fig. 2 shows the system block diagram. The system is connected to the internet through the MODEM in order to access the files being sent through cloud through the web application. The site's desktop computer sends the file for printing. Upon completion of the printing process, the microcontroller will activate the signal to the robotic arm to deliver the print-out to the storage bin. Sensors ensure that the robotic arm moves in its precise direction.

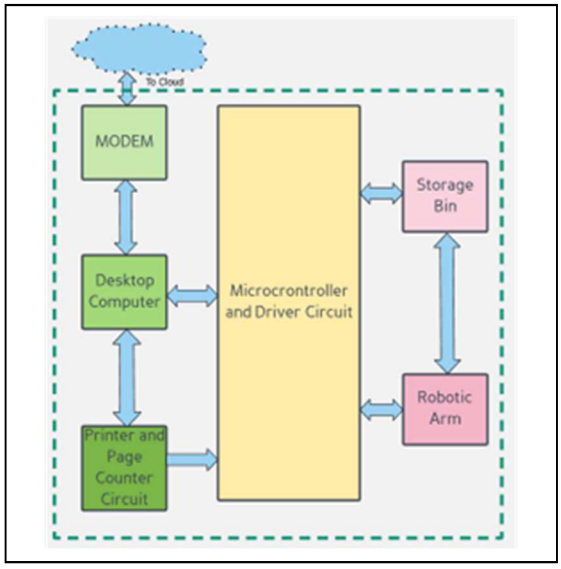



Fig. 2. System Block Diagram

Table 1 shows the major design specifications. The entire system will be controlled through ATMEGA328P-PU and shall be programmed using the Arduino IDE. It shall be integrated to the main Python program that also controls the behavior of the files sent to the cloud. The system consists of a combination of linear and revolute axes, designed to four degrees of freedom. There are four LASER sensors and four Limit switches that will provide sequence and timing. Various motors are used to provide motion for each predefined axis.

TABLE I. SYSTEM SPECIFICATIONS

| Criteria | Description |
|---|---|
| Over-all Size | Over-all required space: 83" x 40 " x 57"<br>Robotic Arm required space: 60" x 40" x 48"<br>Robotic Arm length measurements:<br>Linear X actuator – 40"<br>Linear Z Actuator – 46"<br>Arm Base – 18" x 18" x 3" |
| Power Supply Requirements | 115 – 230 V AC, 6-10 A<br>50-60 Hz<br>5V for the microcontroller |
| Control Mechanism | ATMEGA328P-PU<br>Details: ATMega328 AVR Microcontroller as used in Arduino Uno Boards |
| Degrees of Freedom | Robotic Arm: Four, two linear and two revolute<br>Linear X, Revolute Y, Linear Z, Roll<br>Machine: 1 Revolute Motion for Lock/Unlock |
| Control System Type | Linear Proportional Feedback Control<br>Multiple Input- Multiple Output System (MIMO) |
| Sensors | 4 3V LASER Sensors |

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| | |
|---|---|
| | 4 Limit Switches as contact sensors |
| Actuators | Electric Motors<br>Robotic Arm:<br>Long Screw Bolt for Linear Z<br>Power Window Motor with Gear for Linear X<br>Four Wheels and Axle for Revolute Y<br>Power Window Motor for Roll<br>Storage Compartment System:<br>DC motor with Gear Reducer for Lock-Unlock |
| Printer | HP Deskjet 1112 |
| GUI Type | Browser-Based |

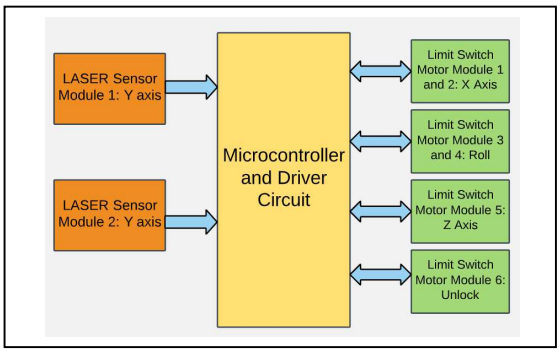



Fig. 3. System Block Diagram

Fig. 3 shows the block diagram of the microcontroller and sensor circuit. Five switch modules support the robotic arm X, Z axes and the roll motors. Another switch module support the lock-unlock revolute motion located on the side of the storage bin in order to release and secure the compartment tray to and from the robotic arm.

To ensure the accurate timing and position of motor action in every axis, iterations where done on the daily function of the Arduino IDE program. First, a careful approximation of delay per axis were encoded in the delay() function. The program is compiled for errors and uploaded in the microcontroller. The Python main program is enabled. This is where hardware connectivity and motor response from sensors were checked. Lastly, an evaluation is done if the delay inputs were sufficient or shall be subject to change in the next iteration.

## IV. RESULTS AND DISCUSSION

The state of each sensor is checked in order to ensure the system's appropriate response. Table 2 shows the sensor condition of each sub-block and its corresponding axis. All the switches are turned off during no-contact and turned on during contact which is the expected behavior.

TABLE II. CONTACT SWITCH TESTING

| SW# | Sub-block | Function | No-contact state | Contact state |
|---|---|---|---|---|
| 1 | Robotic Arm | Linear X | OFF | ON |
| 2 | Robotic Arm | Linear X | OFF | ON |
| 3 | Robotic Arm | Roll | OFF | ON |
| 4 | Robotic Arm | Roll | OFF | ON |
| 5 | Robotic Arm | Linear Z | OFF | ON |
| 6 | Storage Bin | Unlock | OFF | ON |

### *C. OFFSITE PERFORMANCE*

Phase 1 – Compartment Reservation: Service Time 1 min and 43 sec

Table 3 shows the response time of the motors in every stage of Phase 1 of offsite performance. Service time is one minute and 43 seconds. Note that the remote subscriber should wait for this phase before printing.

Table III. OFFSITE PERFORMANCE PHASE 1- COMPARTMENT RESERVATION

| Movement | Definition | Response Time (sec) |
|---|---|---|
| 1. Initial Position | | |
| 2. Online Signal | Towards Reservation of Compartment | 6 |
| 2. z-- | Towards Reservation of Compartment (z=0) | 12 |
| 3. z-- | Towards Reservation of Compartment | 23 |
| 4. x++ | Towards Reservation of Compartment | 22 |
| 5. Unlock | Unlock, Arm will get one compartment | 10 |
| 6. x-- | Arm delivers compartment near printer | 10 |
| 7. z++ | Arm delivers compartment near printer | 17 |
| 8. x++ | Arm delivers compartment | 9 |

Phase 2 –Remote Printing: Service Time 2 min and 7 sec

Table 4 shows the response and motor action time in Phase 2 of the system offsite performance. Service time is 2 min and 7 sec excluding the Cloud printing application which usually takes on the average of 1 min depending on the user. Note that the remote subscriber does not need to wait for this time since this does not require human signaling.

TABLE 4 OFFSITE PERFORMANCE PHASE 2 - REMOTE PRINTING

| Movement | Definition | Response Time (in seconds) |
|---|---|---|
| 1. Previous Position | | |
| 2. File Printing | Time will start after click of the print button and ends upon complete print-out of a two-page colored-picture, test file | 15 |
| 3. z-- | Back to initial position | 13 |
| 4.z++ | Placing of print-out to compartment | 20 |
| 5. x++ | Placing of print-out to compartment | 7 |

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| | | |
|---|---|---|
| 6. Lock | Secure Compartment in Place. Leave compartment in the storage | 13 |
| 7. x-- | Leave compartment in the storage. Arm back in the initial position | 45 |
| 8. z++ | Arm back in the initial position | 14 |

### *C. ONSITE PERFORMANCE*

Compartment No. 2: Service Time 3 min and 3 sec

Table 5 shows the response time of the motors in every stage of onsite performance. Service time is 3 minutes and 3 seconds. This does not include the login time and the retrieval signal which usually takes 30 seconds. Note that the movements highlighted in red are the stages that the subscriber must wait in order to get the print-out. Hence, user may leave the place after stage 8.

TABLE 5 ONSITE PERFORMANCE

| Movement | Definition | Response Time (in seconds) |
|---|---|---|
| 1. Initial Position | | 11 |
| 3. z-- | Towards release of print-out to user | 16 |
| 4. z++ | Towards release of print-out to user | 14 |
| 4. x++ | Towards release of print-out to user | 15 |
| 5. Unlock | Towards release of print-out to user | 14 |
| 6. x-- | Towards release of print-out to user | 6 |
| 7. z++ | Towards release of print-out to user | 7 |
| 8. y++ | Counterclockwise movement towards user retrieval of print-out | 5 |
| 9. y-- | Counterclockwise movement after user retrieval | 5 |
| 10. z-- | Return of compartment to sorting bin | 24 |
| 11. z++ | Return of compartment to sorting bin | 10 |
| 12. x++ | Return of compartment to sorting bin | 13 |
| 12. Lock | Back to initial state | 14 |
| 13. x-- | Back to initial state | 7 |
| 14. z++ | Back to initial state | 15 |
| 15. x++ | Back to initial state | 12 |

Table 6 shows every section of the system including the portions that are affected by human factors. Note that while there is no absolute time that can represent every phase, the sample time was taken as a service for two-page print-out with pictures assigned at compartment number 2. Variation of time is narrow for all the system-controlled phases and is intermediate to wide on all user-controlled phases.

TABLE 6 SYSTEM PHASE

| System Phase | User/System Controlled? | Actual Time/Est. Average Time | Time Variation | Cause of Variation |
|---|---|---|---|---|
| Offsite Performance: Compartment Reservation | System | 1 min 43 sec | Narrow | Compartment slot |
| Cloud Printing | User | 20 sec – 10 min | Wide | Page numbers, print-out contents, User familiarity |
| Offsite Performance: Remote Printing | System | 2 min 7 sec | Narrow | Compartment slot |
| Travel Time | User | 1 min – 2 hrs. | Wide | User to site distance |
| User Login and Retrieval Request | User | 20 sec | Intermediate | User familiarity |
| Onsite Performance | User | 3 min 3 sec | Narrow | Compartment slot |

### *D. USER EVALUATION*

Table 8 shows the summary of the computed survey results. The respondents, on the average, puts most of the criterion on the 3.51-4.50 range (Description: Agree). Highest scores obtained on the system being Operable, Attractive and Can Be Easily Analyzed. Only one falls within the Neutral Range and that is the system being Fault Tolerant. On the other hand, the standard deviation scores show that only the system being Suitable, Attractive and Maintainability Compliant has a consistent rating while the rest are neutral with the system being Not-time Consuming being the only inconsistent.

TABLE 8 USER EVALUATION

| Criterion | Ave Score ($\bar{x}$) | Remarks ($\bar{x}$) | Standard Deviation ($\sigma$) | Remarks ($\sigma$) |
|---|---|---|---|---|
| 1. Suitable | 4.25 | Agree | 0.46291 | Consistent |
| 2. Accurate | 3.875 | Agree | 0.64087 | Neutral |
| 3. Inter-operable | 4 | Agree | 0.755929 | Neutral |
| 4. Secure | 4.125 | Agree | 0.834523 | Neutral |
| 5. Functionally Compliant | 3.875 | Agree | 0.64087 | Neutral |
| 6. Mature | 3.875 | Agree | 0.991031 | Neutral |

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| | | | | |
|---|---|---|---|---|
| 7. Fault Tolerant | 3.5 | Neutral | 0.92582 | Neutral |
| 8. Recoverable | 4 | Agree | 0.755929 | Neutral |
| 9. Reliability Compliant | 4 | Agree | 0.534522 | Neutral |
| 10. Understandable | 3.875 | Agree | 0.834523 | Neutral |
| 11. Can be Easily Learned | 4.5 | Agree | 0.534522 | Neutral |
| 12. Operable | 4.375 | Agree | 0.517549 | Neutral |
| 13. Attractive | 4.375 | Agree | 0.46291 | Consistent |
| 14. Usability Compliant | 4 | Agree | 0.534522 | Neutral |
| 15. Not time-consuming | 3.75 | Agree | 1.035098 | Inconsistent |
| 16. Effective Use of Resources | 4.125 | Agree | 0.834523 | Neutral |
| 17. Efficiency Compliant | 4 | Agree | 0.755929 | Neutral |
| 18. Can be Easily Analyzed | 4.375 | Agree | 0.755929 | Neutral |
| 19. Can be Changed | 4.25 | Agree | 0.517549 | Neutral |
| 20. Stable | 4 | Agree | 0.707107 | Neutral |
| 21. Can be tested | 4.125 | Agree | 0.534522 | Neutral |
| 22. Maintainability Compliant | 4.25 | Agree | 0.46291 | Consistent |
| Average | 4.0625 | Agree | 0.46291 | |

## V. CONCLUSION

The study was able to use Cloud Printing Systems to design and develop a system of accepting, storing and releasing of files through the Automatic Sorting Bin. The study was able to do design a Microcontroller circuit using Arduino as the system control mechanism to enable precise and systematic operation. Moreover, the study was able to successfully incorporate the design and construction of the robotic arm using motors, LASER sensors and limit switches that enables collection and release of the printouts to the user. The automated sorting bin was also made through a tray-type system of storage. The robotic arm provides the proper movement of printouts from the printer to the storage, release and return to its original position. The study incorporates the web application to communicate from the cloud system to printer

The design can be improved in terms of prototype size, speed and job order capacity with the help of better microcontrollers and ergonomics. The system can also be explored to cater other storage and delivery mechanisms aside from paper printouts. Also use of universal mass printing management software can extend the capability of the system to a comprehensive, location-based printing.

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